

Relearning Mixtures of Gaussians

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Abstract

We provide an algorithm for properly learning mixtures of two single-dimensional Gaussians without any separability assumptions. Given $\tilde{O}(1/\varepsilon^2)$ samples from an unknown mixture, our algorithm outputs a mixture that is ε -close in total variation distance, in time $\tilde{O}(1/\varepsilon^{7.01})$. Our sample complexity is optimal up to logarithmic factors, and significantly improves upon both Kalai et al [KMV10], whose algorithm has a prohibitive dependence on $1/\varepsilon$, and Feldman et al. [FOS06], whose algorithm requires bounds on the mixture parameters and depends pseudo-polynomially in these parameters.

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1 Introduction

Learning mixtures of Gaussian distributions is one of the most fundamental problems in Statistics, with a multitude of applications in the natural and social sciences, which has recently received considerable attention in Computer Science literature. Given independent samples from an unknown mixture of Gaussians, the task is to ‘learn’ the underlying mixture.

In one version of the problem, ‘learning’ means estimating the *parameters* of the mixture, that is the mixing probabilities as well as the parameters of each constituent Gaussian. The most popular heuristic for doing so is running the EM algorithm on samples from the mixture [DLR77], albeit no rigorous guarantees are known for it in general.

A line of research initiated by Dasgupta [Das99, AK01, VW02, AM05, BV08] provides rigorous guarantees under separability conditions: roughly speaking, it is assumed that the constituent Gaussians have variation distance bounded away from 0 (indeed, in some cases, distance exponentially close to 1). This line of work was recently settled by a triplet of breakthrough results [KMV10, MV10, BS10], establishing the polynomial solvability of the problem under minimal separability conditions for the parameters to be recoverable in the first place: for any $\varepsilon > 0$, polynomial in n and $1/\varepsilon$ samples from a mixture of n -dimensional Gaussians suffice to recover the parameters of the mixture in $\text{poly}(n, 1/\varepsilon)$ time.

While these results settle the polynomial solvability of the problem, they serve more as a proof of concept in that their dependence on $1/\varepsilon$ is quite expensive.¹ Indeed, even for mixtures of two single-dimensional Gaussians, a practically efficient algorithm for this problem is unknown.

A weaker goal for the learner of Gaussian mixtures is this: given samples from an unknown mixture, find *any* mixture that is close to the unknown one, for some notion of closeness. This PAC-style version of the problem [KMR⁺94] was pursued by Feldman et al. [FOS06] who obtained efficient learning algorithms for mixtures of n -dimensional, axis-aligned Gaussians. Given $\text{poly}(n, 1/\varepsilon, L)$ samples from such mixture, their algorithm constructs a mixture whose KL divergence to the sampled one is at most ε . Unfortunately, the sample and time complexity of their algorithm depends polynomially on a (priorly known) bound L , determining the range of the means and variances of the constituent Gaussians in every dimension.² In particular, the algorithm has pseudo-polynomial dependence on L where there shouldn’t be any dependence on L at all [KMV10, MV10, BS10].

Finally, yet a weaker goal for the learner would be to construct *any distribution* that is close to the unknown mixture. In this *non-proper* version of the problem the learner is not restricted to output a Gaussian mixture, but can output any (representation of a) distribution that is close to the unknown mixture. For this problem, recent results of Diakonikolas et al. [CDSS] provide algorithms for single-dimensional mixtures, whose sample complexity has near-optimal dependence on $1/\varepsilon$. Namely, given $\tilde{O}(1/\varepsilon^2)$ samples from a single-dimensional mixture, they construct a piecewise polynomial distribution that is ε -close in total variation distance.

Inspired by this recent progress on non-properly learning single-dimensional mixtures, our goal in this paper is to provide sample-optimal algorithms that *properly learn*. We obtain such algorithms for mixtures of two single-dimensional Gaussians. Namely,

¹For example, the single-dimensional algorithm in the heart of [KMV10] has sample and time complexity of $\Theta(1/\varepsilon^{300})$ and $\Omega(1/\varepsilon^{1377})$ respectively (even though the authors most certainly did not intend to optimize their constants).

²In particular, it is assumed that every constituent Gaussian in every dimension has mean $\mu \in [-\mu_{\max}, \mu_{\max}]$ and variance $\sigma^2 \in [\sigma_{\min}^2, \sigma_{\max}^2]$ where $\mu_{\max}\sigma_{\max}/\sigma_{\min} \leq L$.

Theorem 1. *For all $\varepsilon, \delta > 0$, given $\log(1/\delta) \cdot \tilde{O}(1/\varepsilon^2)$ independent samples from an arbitrary mixture F of two univariate Gaussians we can compute in time $\log(1/\delta) \cdot O(1/\varepsilon^{7.01})$ a mixture F' such that $d_{TV}(F, F') \leq \varepsilon$ with probability at least $1 - \delta$.*

Discussion. Note that our algorithm makes no separability assumptions about the constituent Gaussians of the unknown mixture, nor does it require or depend on a bound on the mixture's parameters. Moreover, there is nothing special about the constant 7.01 in the exponent of the running time and this constant can be replaced by any constant larger than 7. Finally, it is easy to see that our sample complexity is optimal up to logarithmic factors. Indeed, a Gaussian mixture can trivially simulate a Bernoulli distribution as follows. Let Z be a Bernoulli random variable that is 0 with probability $1 - p$ and 1 with probability p . Clearly, Z can be viewed as a mixture of two Gaussian random variables of 0 variance, which have means 0 and 1 and are mixed with probabilities $1 - p$ and p respectively. It is known that $1/\varepsilon^2$ samples are needed to properly learn a Bernoulli distribution, hence this lower bound immediately carries over to Gaussian mixtures.

Comparison to prior work. In comparison to the recent breakthrough results [KMV10, MV10, BS10], our algorithm has near-optimal sample complexity and milder running time, whereas these results have quite expensive dependence of both their sample and time complexity on the accuracy ε , even for single-dimensional mixtures.³ On the other hand, our algorithm has weaker guarantees in that we properly learn but don't do parameter estimation. In comparison to [FOS06], our algorithm requires no bounds on the parameters of the constituent Gaussians and exhibits no pseudo-polynomial dependence of the sample and time complexity on such bounds. On the other hand, we learn with respect to the total variation distance rather than the KL divergence. Finally, compared to [CDSS13, CDSS], we properly learn while they non-properly learn and we both have near-optimal sample complexity.

Approach. Our algorithm is intuitively quite simple, although some care is required to make the ideas work. First, we can guess the mixing weight up to additive error $O(\varepsilon)$, and proceed with our algorithm pretending that our guess is correct. Every guess will result in a collection of candidate distributions, and the final step of our algorithm is a tournament that will select, from among all candidate distributions produced in the course of our algorithm, a distribution that is ε -close to the unknown mixture, if such a distribution exists. Indeed, devising such a tournament (Theorem 18) is one of our main contributions, and it requires strengthening the tournament-based approach described by Devroye and Lugosi in Chapter 6 of [DL01] (see also [Yat85, DL96, DL97] as well as the recent papers of Daskalakis et al. [DDS12] and Chan et al. [CDSS13]) to continuous distributions whose crossings are difficult to compute exactly, as well as improving the running time to almost linear in the number of candidate distributions. It is worth noting that the tournament based approach of [FOS06] is not applicable here as it requires a priori known bound on the mixture's parameters and depends pseudopolynomially on this bound.

Tuning the number of samples according to the guessed mixing weight, we proceed to draw further samples from the unknown mixture, expecting that some of these samples will fall sufficiently close to the means of the constituent Gaussians, where the closeness will depend on the number of samples drawn as well as the unknown variances. Again we guess which pair of samples fall close

³For example, compared to [KMV10] we improve by a factor of at least 150 the exponent of both the sample and time complexity.

to the means of our constituent Gaussians, at which point we have our hands on 3 out of the 5 parameters of the mixture. To pin down the variances of the constituent Gaussians, we implement a natural idea. Intuitively, if we draw samples from the mixture, we expect that the constituent Gaussian with the smallest σ/w (standard deviation to mixing weight ratio) will determine the smallest distance among the samples, while the one with the largest ratio will determine either the largest sample or the smallest sample. This should be true unless the σ/w ratios are close. Pursuing this idea we produce a collection of variance pairs, one of which truly corresponds to the variances of the constituent Gaussians.

We show that, for any target total variation distance ε , the number of samples required to execute the steps outlined above in order to produce a collection of candidate hypotheses one of which is ε -close to the unknown mixture, as well as to run the tournament to select from among the candidate distributions are $\tilde{O}(1/\varepsilon^2)$. The running time is $\tilde{O}(1/\varepsilon^{7+\gamma})$, for any $\gamma > 0$.

2 Preliminaries

Let $\mathcal{N}(\mu, \sigma^2)$ represent the univariate normal distribution, with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 \in \mathbb{R}$, with density function

$$\mathcal{N}(\mu, \sigma^2, x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

The univariate half-normal distribution with parameter σ^2 is the distribution of $|Y|$ where Y is distributed according to $\mathcal{N}(0, \sigma^2)$. The CDF of the half-normal distribution is

$$F(\sigma, x) = \text{erf}\left(\frac{x}{\sigma\sqrt{2}}\right),$$

where $\text{erf}(x)$ is the error function, defined as

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

We also make use of the complement of the error function, $\text{erfc}(x)$, defined as $\text{erfc}(x) = 1 - \text{erf}(x)$.

A Gaussian mixture model (GMM) of distributions $\mathcal{N}_1(\mu_1, \sigma_1^2), \dots, \mathcal{N}_n(\mu_n, \sigma_n^2)$ has PDF

$$f(x) = \sum_{i=1}^n w_i \mathcal{N}(\mu_i, \sigma_i^2, x),$$

where $\sum_i w_i = 1$. These w_i are referred to as the mixing weights. Drawing a sample from a GMM can be visualized as the following process: select a single Gaussian, where the probability of selecting a Gaussian is equal to its mixing weight, and draw a sample from that Gaussian. In this paper, we consider mixtures of two Gaussians, so $w_2 = 1 - w_1$. We will interchangeably use w and $1 - w$ in place of w_1 and w_2 .

The total variation distance between two probability measures P and Q on a σ -algebra F is defined by

$$d_{\text{TV}}(P, Q) = \sup_{A \in F} |P(A) - Q(A)| = \frac{1}{2} \|P - Q\|_1.$$

For simplicity in the exposition of our algorithm, we make the standard assumption (see, e.g., [FOS06, KMV10]) of infinite precision real arithmetic. In particular, the samples we draw from a mixture of Gaussians are real numbers, and we can do exact computations on real numbers, e.g., we can exactly evaluate the PDF of a Gaussian distribution on a real number.

2.1 Bounds on Total Variation Distance for GMMs

We recall a result from [DDO⁺13]:

Proposition 2 (Proposition B.4 of [DDO⁺13]). *Let $\mu_1, \mu_2 \in \mathbb{R}$ and $0 \leq \sigma_1 \leq \sigma_2$. Then*

$$d_{\text{TV}}(\mathcal{N}(\mu_1, \sigma_1^2), \mathcal{N}(\mu_2, \sigma_2^2)) \leq \frac{1}{2} \left(\frac{|\mu_1 - \mu_2|}{\sigma_1} + \frac{\sigma_2^2 - \sigma_1^2}{\sigma_1^2} \right).$$

The following proposition, whose proof is deferred to the appendix, provides a bound on the total variation distance between two GMMs in terms of the distance between the constituent Gaussians.

Proposition 3. *Suppose we have two GMMs X and Y , with PDFs $w\mathcal{N}_1 + (1-w)\mathcal{N}_2$ and $\hat{w}\hat{\mathcal{N}}_1 + (1-\hat{w})\hat{\mathcal{N}}_2$ respectively. Then $d_{\text{TV}}(X, Y) \leq |w - \hat{w}| + wd_{\text{TV}}(\mathcal{N}_1, \hat{\mathcal{N}}_1) + (1-w)d_{\text{TV}}(\mathcal{N}_2, \hat{\mathcal{N}}_2)$.*

Combining these propositions, we obtain the following lemma:

Lemma 4. *Let X and Y be two GMMs with PDFs $w_1\mathcal{N}_1 + w_2\mathcal{N}_2$ and $\hat{w}_1\hat{\mathcal{N}}_1 + \hat{w}_2\hat{\mathcal{N}}_2$ respectively, where $|w_i - \hat{w}_i| \leq \frac{\varepsilon}{3}$, $|\mu_i - \hat{\mu}_i| \leq \frac{\varepsilon}{4}\sigma_i$, $|\sigma_i - \hat{\sigma}_i| \leq \frac{\varepsilon}{8}\sigma_i$, for all i such that $w_i \geq \frac{\varepsilon}{3}$. Then $d_{\text{TV}}(X, Y) \leq \varepsilon$.*

2.2 Outline of the Algorithm

We can decompose our algorithm into two components: generating a collection of candidate distributions containing at least one candidate with low statistical distance to the unknown distribution (Theorem 5), and identifying such a candidate from this collection (Theorem 18).

Generation of Candidate Distributions: In Section 3, we deal with generation of candidate distributions. A *candidate distribution* is described by the parameter set $(\hat{w}, \hat{\mu}_1, \hat{\sigma}_1, \hat{\mu}_2, \hat{\sigma}_2)$, which corresponds to the GMM with PDF $f(x) = \hat{w}\mathcal{N}(\hat{\mu}_1, \hat{\sigma}_1^2, x) + (1 - \hat{w})\mathcal{N}(\hat{\mu}_2, \hat{\sigma}_2^2, x)$. As suggested by Lemma 4, if we have a candidate distribution with sufficiently accurate parameters, it will have low statistical distance to the unknown distribution. Our first goal will be to generate a collection of candidates that contains at least one such candidate. Since the time complexity of our algorithm depends on the size of this collection, we wish to keep it to a minimum.

At a high level, we sequentially generate candidates for each parameter. In particular, we start by generating candidates for the mixing weight. While most of these will be inaccurate, we will guarantee to produce at least one appropriately accurate candidate $\hat{w}^*$. Then, for each candidate mixing weight, we will generate candidates for the means of the two Gaussians. We will guarantee that, out of the candidate means we generated for $\hat{w}^*$, it is likely that at least one pair $(\hat{\mu}_1^*, \hat{\mu}_2^*)$ will be sufficiently close to the true means. The candidate means that were generated for other mixing weights have no such guarantee. We use a similar sequential approach to generate the two variances. We can not immediately identify which candidates have inaccurate parameters, and they serve only to inflate the size of our collection.

At a lower level, our algorithm starts by generating candidates for the mixing weight followed by generating candidates for the means. We observe that if we take n samples from a single Gaussian, it is likely that there will exist a point at distance $O(\frac{\sigma}{n})$ from its mean. Thus, if we take $\Theta(\frac{1}{w_i\varepsilon})$ samples from the mixture, one of them will be sufficiently close to the mean of the corresponding Gaussian. Exploiting this observation we obtain candidates for the mixing weight and the two means as summarized by Lemma 8.

Next, we generate candidates for the variance of the constituent Gaussian with the smaller value of $\frac{\sigma_i}{w_i}$. Note that we do not know which of the two Gaussians this is. The solution is to branch our

algorithm, where each branch assumes a correspondence to a different Gaussian. One of the two branches is guaranteed to be correct, and it will only double the number of candidate distributions. Our specific approach is based on the observation that given n samples from a single Gaussian, the minimum distance of a sample to the mean will likely be $\Theta(\frac{\sigma}{n})$. In the mixture, this property will still hold for the Gaussian with the smaller $\frac{\sigma_i}{w_i}$, so we extract this statistic and use a grid around it to generate sufficiently accurate candidates for σ_i . This is Lemma 12.

Finally, we generate candidates for the remaining variance. Again, we branch our algorithm depending on whether or not the remaining variance is not much larger than the first. In the first case, we simply grid over an interval defined by two multiples of the first variance. Otherwise, one of the extreme samples will be generated from the corresponding Gaussian. By proving bounds on the most extreme of n samples from a Gaussian, we can grid over another interval and obtain candidates for the final parameter. This is Lemma 17.

Putting everything together, we obtain the following result whose proof is in Section 3.5.

Theorem 5. *For all $\varepsilon, \delta > 0$, given $\log(1/\delta) \cdot O(1/\varepsilon^2)$ independent samples from an arbitrary mixture F of two univariate Gaussians, we can generate a collection of $\log(1/\delta) \cdot \tilde{O}(1/\varepsilon^5)$ candidate mixtures of two univariate Gaussians, containing at least one candidate F' such that $d_{TV}(F, F') < \varepsilon$ with probability at least $1 - \delta$.*

Candidate Selection: In view of Theorem 5, to prove our main result it suffices to select from among the candidate mixtures some mixture that is close to the unknown mixture. In Section 4, we describe a tournament-based algorithm (Theorem 18) for identifying a candidate which has low statistical distance to the unknown mixture, concluding the proof of Theorem 1. See our discussion in Section 4 for the challenges arising in obtaining a tournament-based algorithm for continuous distributions whose crossings are difficult to compute exactly, as well as in speeding up the tournament's running time to almost linear in the number of candidates.

3 Generating Candidate Distributions

By Proposition 3, if one of the Gaussians of a mixture has a negligible mixing weight, it has a negligible impact on the mixture's statistical distance to the unknown mixture. Hence, the candidate means and variances of this Gaussian are irrelevant. This is fortunate, since if $\min(w, 1 - w) < \varepsilon$ and we only draw $O(1/\varepsilon^2)$ samples from the unknown mixture, as we are planning to do, we have no hope of seeing a sufficient number of samples from the low-weight Gaussian to perform accurate statistical tests for it. So for this section we will assume that $\min(w, 1 - w) \geq \Omega(\varepsilon)$ and we will deal with the other case separately.

3.1 Generating Mixing Weight Candidates

The first step is to generate candidates for the mean. We can obtain a collection of $O(\frac{1}{\varepsilon})$ candidates containing some $\hat{w} \in [w - \varepsilon, w + \varepsilon]$ by simply taking the set $\{k\varepsilon \mid k \in [\frac{1}{\varepsilon}]\}$.

3.2 Generating Means Candidates

The next step is to generate candidates for the two means. In particular, we want a collection of candidates containing statistics $\hat{\mu}_i$ such that $\mu_i - \varepsilon\sigma_i \leq \hat{\mu}_i \leq \mu_i + \varepsilon\sigma_i$, for $i = 1, 2$. The following propositions are straightforward and proved in Appendix D.

Proposition 6. Fix $i \in \{1, 2\}$. Given $\frac{5\sqrt{\pi/2}}{w_i\varepsilon}$ samples from a GMM, there will exist a sample $\hat{\mu}_i \in \mu_i \pm \varepsilon\sigma_i$ with probability $\geq \frac{99}{100}$.

Proposition 7. Fix $i \in \{1, 2\}$. Suppose $w_i - \varepsilon \leq \hat{w}_i \leq w_i + \varepsilon$, and $w_i \geq \varepsilon$. Then $\frac{2}{\hat{w}_i} \geq \frac{1}{w_i}$.

We use these facts to design a simple algorithm: for each candidate $(\hat{w}_1, \hat{w}_2)$ (from Section 3.1), take $\frac{10\sqrt{\pi/2}}{\hat{w}_1\varepsilon}$ samples from the mixture as candidates for μ_1 and $\frac{10\sqrt{\pi/2}}{\hat{w}_2\varepsilon}$ samples as candidates for μ_2 , and output every combination of them as candidates for (μ_1, μ_2) . To avoid exploding the number of samples, we actually *first* draw a set $\mathcal{S}$ of $\frac{20\sqrt{\pi/2}}{\varepsilon^2}$ samples from the mixture, and *then* for every candidate $(\hat{w}_1, \hat{w}_2)$ we take $\frac{10\sqrt{\pi/2}}{\hat{w}_1\varepsilon}$ samples from $\mathcal{S}$ as candidates for μ_1 and a disjoint $\frac{10\sqrt{\pi/2}}{\hat{w}_2\varepsilon}$ samples from $\mathcal{S}$ as candidates from μ_2 .

We now examine how many candidate triples $(\hat{w}, \hat{\mu}_1, \hat{\mu}_2)$ we generated. Naively, since $\hat{w}_i$ may be as small as $O(\varepsilon)$, the candidates for each mean will multiply the size of our collection by $O(\frac{1}{\varepsilon^2})$. However, we note that when $\hat{w}_i = \Omega(1)$, then the number of candidates for μ_i is actually $O(\frac{1}{\varepsilon})$. We count the number of candidate triples $(\hat{w}, \hat{\mu}_1, \hat{\mu}_2)$, combining with previous results in the following:

Lemma 8. Suppose we have sample access to a GMM with (unknown) parameters $(w, \mu_1, \mu_2, \sigma_1, \sigma_2)$. Then for any $\varepsilon > 0$ and constants $c_w, c_m > 0$, using $O(\frac{1}{\varepsilon^2})$ samples from the GMM, we can generate a collection of $O\left(\frac{\log \varepsilon^{-1}}{\varepsilon^3}\right)$ candidate triples for (w, μ_1, μ_2) . With probability $\geq \frac{49}{50}$, this will contain a triple $(\hat{w}, \hat{\mu}_1, \hat{\mu}_2)$ such that $\hat{w} \in w \pm c_w\varepsilon$, $\hat{\mu}_1 \in \mu_1 \pm c_m\varepsilon\sigma_1$, and $\hat{\mu}_2 \in \mu_2 \pm c_m\varepsilon\sigma_2$.

The proof of the lemma is deferred to App. D. It implies that we can generate $O\left(\frac{\log \frac{1}{\varepsilon}}{\varepsilon^3}\right)$ candidate triples, such that at least one triple simultaneously describes w , μ_1 , and μ_2 to the desired accuracy.

3.3 Generating Candidates for a Single Variance

In this section, we generate candidates for the variance corresponding to the Gaussian with the smaller value of $\frac{\sigma_i}{w_i}$. Note that, a priori, we do not know whether $i = 1$ or $i = 2$. We try both cases, first generating candidates assuming they correspond to σ_1 , and then repeating with σ_2 . This will multiply our total number of candidate distributions by a factor of 2.

So, without loss of essential generality, assume for this section that $i = 1$. The basic idea is that we will find the closest point to μ_1 . We use the following property (whose proof is deferred to Appendix D) to estimate this distance:

Proposition 9. Suppose we have $n > 0$ IID random variables, $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$. Let $Y = \min_i |X_i - \mu|$. Then $\frac{c_1\sigma}{n} \leq Y \leq \frac{c_2\sigma}{n}$ for some constants c_1, c_2 with probability $\geq \frac{9}{10}$.

When sampling a GMM, we can bound the distances from $\hat{\mu}_1$ of the samples coming from the “correct” Gaussian as well as the “incorrect” Gaussian in the following propositions:

Proposition 10. Suppose $\hat{\mu}_1$, our estimate for μ_1 , has the property that $|\hat{\mu}_1 - \mu_1| \leq c_0\varepsilon\sigma_1$. Given $\frac{48}{w_1\varepsilon}$ samples from the mixture, with probability $\geq \frac{17}{20}$, the nearest sample from $\mathcal{N}_1$ to $\hat{\mu}_1$ is at a distance $\in [(\frac{c_1}{72} - c_0)\varepsilon\sigma_1, (\frac{c_2}{24} + c_0)\varepsilon\sigma_1]$.

Proof of Proposition 10: Using standard Chernoff bounds, with probability at least $1 - 2\exp(-\frac{4}{\varepsilon})$, our number of samples from $\mathcal{N}_1$ is in the range $[\frac{24}{\varepsilon}, \frac{72}{\varepsilon}]$. Suppose this condition holds, along with

the conditions for Proposition 9. By examining the effect of the three uncertainties ($\min_i |X_i - \mu_1|$, the number of samples from $\mathcal{N}_1$, and $|\hat{\mu}_1 - \mu_1|$), we can see the extreme cases result in the range $[(\frac{c_1}{72} - c_0)\varepsilon\sigma_1, (\frac{c_2}{24} + c_0)\varepsilon\sigma_1]$. $\square$

Proposition 11. Assume that $\frac{\sigma_1}{w_1} \leq \frac{\sigma_2}{w_2}$. Given $\frac{48}{w_1\varepsilon}$ samples from the mixture, with probability $\geq \frac{22}{25}$, the nearest sample from $\mathcal{N}_2$ to $\hat{\mu}_1$ is at a distance $\geq c_3\varepsilon\sigma_1$, for some constant c_3 .

The straightforward proof of the above proposition is deferred to Appendix D. Putting the two propositions together we obtain the following lemma whose proof is also in the appendix.

Lemma 12. Suppose we have sample access to a GMM with parameters $(w, \mu_1, \mu_2, \sigma_1, \sigma_2)$, where $\frac{\sigma_1}{w} \leq \frac{\sigma_2}{1-w}$. Furthermore, we have estimates $\hat{w} \in w \pm c_w\varepsilon$, $\hat{\mu}_1 \in \mu_1 \pm c_m\varepsilon\sigma_1$, $\hat{\mu}_2 \in \mu_2 \pm c_m\varepsilon\sigma_2$, where $c_w, c_m > 0$ are constants and c_m is sufficiently small. Then for any $\varepsilon > 0$ and constant $c_s > 0$, using $O(\frac{1}{\varepsilon^2})$ samples from the GMM, we can generate a collection of $O(\frac{1}{\varepsilon})$ candidates for σ_1 . With probability $\geq \frac{7}{10}$, this will contain a candidate $\hat{\sigma}_1$ such that $\hat{\sigma}_1 \in (1 \pm c_s\varepsilon)\sigma_1$.

3.4 Generating Candidates for the Other Variance

Without loss of generality, for this section, assume that the unknown variance is σ_2^2 . We distinguish two cases for it:

3.4.1 Subcase 1: $\frac{w_2}{w_1}\sigma_1 \leq \sigma_2 \leq C_2\sigma_1$

In this case, we simply use a multiplicative grid. From Fact 22, the number of candidates generated is

$$\frac{1}{\log(1+\varepsilon)} \log\left(\frac{C_2 w_1}{C_1 w_2}\right) \leq \frac{1}{\log(1+\varepsilon)} \log\left(\frac{C_2}{C_1 \varepsilon}\right) = O\left(\frac{\log \varepsilon^{-1}}{\varepsilon}\right).$$

3.4.2 Subcase 2: $C_2\sigma_1 \leq \sigma_2$

In this case, we examine the most extreme point: the maximum or the minimum of samples from the mixture. If one of the variances is reasonably larger than the other, then it is likely that the most extreme point is from the corresponding Gaussian.

If $\mu_2 \geq \mu_1$, we examine the maximum sample, otherwise, the minimum. As before, we will try both cases, multiplying the total number of parameter candidates by a factor of 2. In this section, we will analyze the maximum sample; the minimum sample case follows symmetrically.

First, we require a fact about the maximum of several Gaussian random variables, whose proof we defer to Appendix D.

Proposition 13. Let $Y = \max_{1 \leq i \leq n} X_i$, where X_i are IID $\mathcal{N}(\mu, \sigma^2)$ random variables. Then

$$\mu + \sigma \frac{1}{\sqrt{\pi \log 2}} \sqrt{\log n} \leq E[Y] \leq \mu + \sigma \sqrt{2} \sqrt{\log n}.$$

Next, we wish to bound the probability that the maximum of several Gaussian random variables deviates highly from the expected value. We apply a classical result in the concentration of measure of Lipschitz functions of Gaussian random variables.

Proposition 14. [Theorem 3.12 in [MP07]] Let $Y = \max_{1 \leq i \leq n} X_i$, where X_i are IID $\mathcal{N}(\mu, \sigma^2)$ random variables. Then

$$\mathbb{P}[|Y - E[Y]| > k\sigma] \leq 2 \exp\left(\frac{-k^2}{2}\right).$$

We combine these facts into the following result, whose proof is in the appendix.

Proposition 15. Suppose we take $n = \frac{2^{10\pi+4}}{\varepsilon^2}$ samples from the Gaussian mixture $w_1\mathcal{N}(\mu_1, \sigma_1^2) + w_2\mathcal{N}(\mu_2, \sigma_2^2)$. Then the maximum sample from $\mathcal{N}_i$ is in the range

$$\left[\mu_i + \sigma_i \left(\frac{1}{\sqrt{\pi \log 2}} \sqrt{\log \frac{w_i n}{2}} - \sqrt{5} \right), \mu_i + \sigma_i \left(\sqrt{2} \sqrt{\log \frac{3w_i n}{2}} + \sqrt{5} \right) \right]$$

with probability $\geq \frac{4}{5}$.

We examine the effect on the maximum sample from the mixture. The proof is deferred to Appendix D.

Proposition 16. Suppose we take $n = O(\frac{1}{\varepsilon^2})$ samples from the Gaussian mixture $w_1\mathcal{N}(\mu_1, \sigma_1^2) + w_2\mathcal{N}(\mu_2, \sigma_2^2)$, $\mu_2 \geq \mu_1$, and $\sigma_2 \geq C_2\sigma_1$. Then the maximum sample from the mixture is in the range

$$\left[\mu_2 + \sigma_2 \left(\frac{1}{\sqrt{\pi \log 2}} \sqrt{\log \frac{w_2 n}{2}} - \sqrt{5} \right), \mu_2 + \sigma_2 \left(\sqrt{2} \sqrt{\log \frac{3w_2 n}{2}} + \sqrt{5} \right) \right]$$

with probability $\geq \frac{19}{20}$.

Now that we know that the greatest sample is from $\mathcal{N}_2$, we must show this interval is efficiently griddable. We combine this with previous details into the following Lemma, whose proof is deferred to Appendix D.

Lemma 17. Suppose we have sample access to a GMM with parameters $(w, \mu_1, \mu_2, \sigma_1, \sigma_2)$, where $\frac{\sigma_1}{w} \leq \frac{\sigma_2}{1-w}$. Furthermore, we have estimates $\hat{w} \in w \pm c_w \varepsilon$, $\hat{\mu}_1 \in \mu_1 \pm c_m \varepsilon \sigma_1$, $\hat{\mu}_2 \in \mu_2 \pm c_m \varepsilon \sigma_2$, $\hat{\sigma}_1 \in (1 \pm c_s \varepsilon) \sigma_1$, where $c_w, c_m, c_s > 0$ are constants. Then for any $\varepsilon > 0$, using $O(\frac{1}{\varepsilon^2})$ samples from the GMM, we can generate a collection of $O\left(\frac{\log \varepsilon^{-1}}{\varepsilon}\right)$ candidates for σ_2 . With probability $\geq \frac{19}{20}$, this will contain a candidate $\hat{\sigma}_2$ such that $\hat{\sigma}_2 \in (1 \pm c_s \varepsilon) \sigma_2$.

3.5 Putting It Together

Theorem 5 is obtained by combining the previous sections, and its proof is given in the appendix.

4 The Tournament

In view of Theorem 5, to prove Theorem 1 we need to choose from among a collection of candidate distributions a distribution that is close to an unknown mixture. This is achieved by algorithm **FastTournament** provided by Theorem 18. **FastTournament** sets up a tournament among the candidate distributions, making sure that the distribution selected by the tournament is close to the unknown mixture. This basic approach of having candidate hypotheses compete against each other is quite natural and has been commonly used in the literature; see e.g. Devroye and

Lugosi ([DL96, DL97] and Chapter 6 of [DL01]), Yatracos [Yat85]), as well as the recent papers of Daskalakis et al. [DDS12] and Chan et al [CDSS13]. However, these approaches are either slower (having quadratic dependence of their running times on the number of candidate hypotheses), or are not applicable to continuous distributions whose crossings are difficult to compute exactly.

Theorem 18. *For any constant $\gamma > 0$, there is an algorithm $\text{FastTournament}_\gamma^X(\mathcal{H}, \epsilon, \delta)$ which is given oracle access to a mixture of two single-dimensional Gaussians X , a set $\mathcal{H}$ of mixtures of two single-dimensional Gaussians each of which is parametrically described (by providing the mixing weight as well as the means and variances of the constituent Gaussians), an accuracy parameter $\epsilon > 0$, and a confidence parameter $\delta > 0$. The algorithm makes*

$$m = O(\log(|\mathcal{H}|/\delta)/\epsilon^2)$$

draws from X and returns some $H \in \mathcal{H}$ or declares “failure”. If there exists some $H^ \in \mathcal{H}$ such that $d_{\text{TV}}(H^*, X) \leq \epsilon$ then with probability at least $1 - \delta$ the H that FastTournament returns has $d_{\text{TV}}(H, X) \leq O(\epsilon)$. The running time of the algorithm is $O(|\mathcal{H}|^{1+\gamma} \log(|\mathcal{H}|/\delta)/\epsilon^2)$.*

FastTournament is based on a recursive application of a slower tournament-based algorithm, SlowTournament , encapsulated in the following lemma. SlowTournament performs a pairwise comparison between all candidate hypotheses using the subroutine Choose-Hypothesis of Lemma 20 resulting in a quadratic running time. FastTournament organizes the tournament in a more efficient manner thereby improving the running time to almost linear.

Lemma 19. *There is an algorithm $\text{SlowTournament}^X(\mathcal{H}, \epsilon, \delta)$ which is given oracle access to a mixture of two single-dimensional Gaussians X , a set $\mathcal{H}$ of mixtures of two single-dimensional Gaussians each of which is parametrically described, an accuracy parameter $\epsilon > 0$, and a confidence parameter $\delta > 0$. The algorithm makes $m = O(\log(|\mathcal{H}|/\delta)/\epsilon^2)$ draws from X and returns some $H \in \mathcal{H}$ or declares “failure”. If there exists some $H^* \in \mathcal{H}$ such that $d_{\text{TV}}(H^*, X) \leq \epsilon$ then with probability at least $1 - \delta$ the H that SlowTournament returns has $d_{\text{TV}}(H, X) \leq 8\epsilon$. The running time of the algorithm is $O(|\mathcal{H}|^2 \log(|\mathcal{H}|/\delta)/\epsilon^2)$.*

Lemma 20. *There is an algorithm $\text{Choose-Hypothesis}^X(H_1, H_2, \epsilon, \delta)$ which is given oracle access to a mixture X , parametric descriptions of two hypothesis mixtures H_1, H_2 for X , an accuracy parameter ϵ , and a confidence parameter δ . It makes $m = O(\log(1/\delta)/\epsilon^2)$ draws from X and returns some $H \in \{H_1, H_2\}$ as the winner or declares a “draw”. The running time of the algorithm is $O(\log(1/\delta)/\epsilon^2)$. Additionally:*

1. *If $d_{\text{TV}}(X, H_1) \leq \epsilon$ and $d_{\text{TV}}(X, H_2) > 8\epsilon$, the probability that H_1 is not returned is at most δ ;*
2. *If $d_{\text{TV}}(X, H_1) \leq \epsilon$ and $d_{\text{TV}}(X, H_2) > 4\epsilon$, the probability that H_2 is returned is at most δ .*

The analogous conclusions hold if we interchange H_1 and H_2 . Moreover, if $d_{\text{TV}}(H_1, H_2) \leq 5\epsilon$, the algorithm will declare a “draw” with probability at least $1 - \delta$.

Proof of Lemma 20: We set up a competition between H_1 and H_2 . First, we define the following subset of $\mathbb{R}$:

$$\mathcal{W}_1 = \mathcal{W}_1(H_1, H_2) := \{w \in \mathbb{R} \mid H_1(w) > H_2(w)\}. \quad (1)$$

In terms of $\mathcal{W}_1$ we define $p_1 = H_1(\mathcal{W}_1)$ and $p_2 = H_2(\mathcal{W}_1)$. Clearly, $p_1 > p_2$ and $d_{TV}(H_1, H_2) = p_1 - p_2$. The competition between H_1 and H_2 is carried out as follows:

- 1a. Draw $m = O\left(\frac{\log(1/\delta)}{\epsilon^2}\right)$ samples $s_1, \dots, s_m$ from X , and let $\hat{\tau} = \frac{1}{m}|\{i \mid s_i \in \mathcal{W}_1\}|$ be the fraction of them that fall inside $\mathcal{W}_1$.
- 1b. Similarly, draw m samples from H_1 , and let $\hat{p}_1$ be the fraction of them that fall inside $\mathcal{W}_1$.
- 1c. Finally, draw m samples from H_2 , and let $\hat{p}_2$ be the fraction of them that fall inside $\mathcal{W}_1$.
2. If $\hat{p}_1 - \hat{p}_2 \leq 6\epsilon$, declare a draw. Otherwise:
3. If $\hat{\tau} > \hat{p}_1 - 2\epsilon$, declare H_1 as winner and return H_1 ; otherwise,
4. if $\hat{\tau} < \hat{p}_2 + 2\epsilon$, declare H_2 as winner and return H_2 ; otherwise,
5. Declare a draw.

For the execution of our algorithm, notice that we can check whether a sample falls inside $\mathcal{W}_1$ by evaluating the PDF of both H_1 and H_2 on that sample. Moreover, given a parametric description $(w, \mu_1, \sigma_1^2, \mu_2, \sigma_2^2)$ of a mixture, we can draw a sample from it as follows: first draw a uniform $[0, 1]$ variable whose value compared to w determines whether to sample from $\mathcal{N}(\mu_1, \sigma_1^2)$ or $\mathcal{N}(\mu_2, \sigma_2^2)$ in the second step; for the second step, use the Box-Muller transform [BM58] to obtain sample from either $\mathcal{N}(\mu_1, \sigma_1^2)$ or $\mathcal{N}(\mu_2, \sigma_2^2)$ as decided in the first step.

The correctness of **Choose-Hypothesis** is a consequence of the following claim whose proof is deferred to the Appendix.

Claim 1. *Suppose that $d_{TV}(X, H_1) \leq \epsilon$. Then:*

1. *If $d_{TV}(X, H_2) > 8\epsilon$, then the probability that the competition between H_1 and H_2 does not declare H_1 as the winner is at most $6e^{-m\epsilon^2/2}$;*
2. *If $d_{TV}(X, H_2) > 4\epsilon$, then the probability that the competition between H_1 and H_2 returns H_2 as the winner is at most $6e^{-m\epsilon^2/2}$.*

The analogous conclusions hold if we interchange H_1 and H_2 . Finally, if $d_{TV}(H_1, H_2) \leq 5\epsilon$, the algorithm will declare a draw with probability at least $1 - 6e^{-m\epsilon^2/2}$.

□

Proof of Lemma 19: Draw $O(\log(2|\mathcal{H}|/\delta)/\epsilon^2)$ samples from X and, using the same samples, run

$$\text{Choose-Hypothesis}^X\left(H_1, H_2, \epsilon, \frac{\delta}{2|\mathcal{H}|}\right),$$

for every pair of distributions $H_1, H_2 \in \mathcal{H}$. If there exists a distribution $H \in \mathcal{H}$ that was never a loser (but potentially tied with some distributions), output any such distribution. Otherwise, output “failure.”

We analyze the correctness of our proposed algorithm in two steps. First, suppose there exists $H^* \in \mathcal{H}$ such that $d_{TV}(H^*, X) \leq \epsilon$. We argue that, with probability at least $1 - \frac{\delta}{2}$, H^* never loses

a competition against any other $H' \in \mathcal{H}$ (so the tournament does not output “failure”). Consider any $H' \in \mathcal{Q}$. If $d_{TV}(X, H') > 4\epsilon$, by Lemma 20 the probability that H^* is not declared a winner or tie against H' is at most $\frac{\delta}{2N}$. On the other hand, if $d_{TV}(X, H') \leq 4\epsilon$, the triangle inequality gives that $d_{TV}(H^*, H') \leq 5\epsilon$ and, by Lemma 20, the probability that H^* does not draw against H' is at most $\frac{\delta}{2N}$. A union bound over all N distributions in $\mathcal{H}$ shows that with probability at least $1 - \frac{\delta}{2}$, the distribution H^* never loses a competition.

We next argue that with probability at least $1 - \frac{\delta}{2}$, every distribution $H \in \mathcal{H}$ that never loses must be 8ϵ -close to X . Fix a distribution H such that $d_{TV}(X, H) > 8\epsilon$. Lemma 20 implies that H loses to H^* with probability at least $1 - \delta/2N$. A union bound gives that with probability at least $1 - \frac{\delta}{2}$, every distribution H such that $d_{TV}(X, H) > 8\epsilon$ loses some competition.

Thus, with overall probability at least $1 - \delta$, the tournament does not output “failure” and outputs some distribution H such that $d_{TV}(H, X) \leq 8\epsilon$. $\square$

Proof of Theorem 18: Set $N = |\mathcal{H}|$ and, for simplicity, assume that $\sqrt{N}$ is integer. (If not, introduce into $\mathcal{H}$ multiple copies of an arbitrary $H \in \mathcal{H}$ so that $\sqrt{N}$ becomes an integer.) Partition $\mathcal{H}$ into $\sqrt{N}$ subsets, $\mathcal{H} = \mathcal{H}_1 \sqcup \mathcal{H}_2 \sqcup \dots \sqcup \mathcal{H}_{\sqrt{N}}$ and do the following:

1. Set $\delta' = \delta/2$, draw $O(\log(\sqrt{|\mathcal{H}|}/\delta')/\epsilon^2)$ samples from X and, using the same samples, run $\text{SlowTournament}^X(\mathcal{H}_i, \epsilon, \delta')$ from Lemma 19 for each i ;
2. Run $\text{SlowTournament}^X(\mathcal{W}, 8\epsilon, \delta')$, where $\mathcal{W}$ are the distributions output by SlowTournament in the previous step. If $\mathcal{W} = \emptyset$ output “failure”.

Let us call the above algorithm $\text{SlowTournament}^{\otimes 1}(\mathcal{H}, \epsilon, \delta)$, before proceeding to analyze its correctness, sample and time complexity. Suppose there exists a distribution $H \in \mathcal{H}$ such that $d_{TV}(H, X) \leq \epsilon$. W.l.o.g. assume that $H \in \mathcal{H}_1$. Then, from Lemma 19, with probability at least $1 - \delta'$, $\text{SlowTournament}^X(\mathcal{H}_1, \epsilon, \delta')$ will output a distribution H' such that $d_{TV}(H', X) \leq 8\epsilon$. Conditioning on this and applying Lemma 19 again, with conditional probability at least $1 - \delta'$ $\text{SlowTournament}^X(\mathcal{W}, 8\epsilon, \delta')$ will output a distribution H'' such that $d_{TV}(H'', X) \leq 64\epsilon$. So with overall probability at least $1 - \delta$, $\text{SlowTournament}^{\otimes 1}(\mathcal{H}, \epsilon, \delta)$ will output a distribution that is 64ϵ -close to X . The number of samples that the algorithm draws from X is $O(\log(|\mathcal{H}|/\delta)/\epsilon^2)$, and the running time is

$$\sqrt{N} \times O(N \log(|\mathcal{H}|/\delta')/\epsilon^2) + O(N \log(|\mathcal{H}|/\delta')/(8\epsilon)^2) = O(|\mathcal{H}|^{3/2} \log(|\mathcal{H}|/\delta)/\epsilon^2).$$

So, compared to SlowTournament , $\text{SlowTournament}^{\otimes 1}$ has the same sample complexity asymptotics and the same asymptotic guarantee for the distance from X of the output distribution, but the exponent of $|\mathcal{H}|$ in the running time improved from 2 to $3/2$.

For $t = 2, 3, \dots$, define $\text{SlowTournament}^{\otimes t}$ by replacing SlowTournament by $\text{SlowTournament}^{\otimes t-1}$ in the code of $\text{SlowTournament}^{\otimes 1}$. It follows from the same analysis as above that as t increases the exponent of $|\mathcal{H}|$ in the running time gets arbitrarily close to 1. In particular, in one step an exponent of $1 + \alpha$ becomes an exponent of $1 + \alpha/2$. So for some constant t , $\text{SlowTournament}^{\otimes t}$ will satisfy the requirements of the theorem. $\square$

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A Gridding

We will encounter settings where we have bounds L and R on an unknown measure X such that $L \leq X \leq R$, and wish to obtain an estimate $\hat{X}$ such that $(1 - \varepsilon)X \leq \hat{X} \leq (1 + \varepsilon)X$. Gridding is a common technique to generate a list of candidates that is guaranteed to contain such an estimate.

Fact 21. Candidates of the form $L + k\varepsilon L$ define an additive grid with at most $\frac{1}{\varepsilon} \left(\frac{R-L}{L} \right)$ candidates.

Fact 22. Candidates of the form $L(1+\varepsilon)^k$ define a multiplicative grid with at most $\frac{1}{\log(1+\varepsilon)} \log \left(\frac{R}{L} \right)$ candidates.

We also encounter scenarios where we require an additive estimate $X - \varepsilon \leq \hat{X} \leq X + \varepsilon$.

Fact 23. Candidates of the form $L + k\varepsilon$ define an absolute additive grid with $\frac{1}{\varepsilon} (R - L)$ candidates.

B Bounds on Forms of $\operatorname{erfc}(x)$

Since the error function often arises when dealing with Gaussian distributions, we require a few limit properties of this function.

Proposition 24. For all $n \geq c \geq 0$,

$$\left(1 - \frac{c}{n}\right)^c e^{-c} \leq \left(1 - \frac{c}{n}\right)^n \leq e^{-c}$$

While the latter inequality is folklore, the former is less commonly used, so we provide a proof.

Proof. For $t \in [1 - \frac{c}{n}, 1]$, $\frac{1}{t} \leq \frac{1}{1 - \frac{c}{n}}$. Integrating with respect to t over $[1 - \frac{c}{n}, 1]$, we see $-\frac{c}{n-c} \leq \ln(1 - \frac{c}{n})$. Exponentiating both sides, raising to the power of $n - c$, and some rearrangement gives the desired inequality. $\square$

Proposition 25. For all $n \geq c \geq 0$,

$$\left(1 - c_1 \operatorname{erf}\left(\frac{\sqrt{\pi} c_2}{2} \frac{c}{n}\right)\right)^n \leq e^{-c_1 c_2}$$

Proof. Since $f(n) = \left(1 - c_1 \operatorname{erf}\left(\frac{\sqrt{\pi} c_2}{2} \frac{c}{n}\right)\right)^n$ is a monotonically increasing function of n for $n > 0$, it suffices to show that $\lim_{n \rightarrow \infty} f(n) \leq e^{-c_1 c_2}$. Taking the Taylor expansion of $\operatorname{erf}(x)$ gives us $f(n) \leq (1 - \frac{c_1 c_2}{n} + O(\frac{1}{n^3}))^n$, which approaches $e^{-c_1 c_2}$ in the limit. $\square$

Proposition 26. For all $n \geq c \geq 0$,

$$\left(1 - \frac{c}{n}\right)^n e^{-c} \leq \operatorname{erfc}\left(\frac{\sqrt{\pi} c}{2} \frac{c}{n}\right)^n$$

Proof. The Taylor expansion of $\operatorname{erfc}(x)$ at $x = 0$ is $1 - \frac{2x}{\sqrt{\pi}} + O(x^3) > 1 - \frac{2x}{\sqrt{\pi}}$ for $x > 0$. This gives us the inequality $1 - \frac{c}{n} \leq \operatorname{erfc}\left(\frac{\sqrt{\pi} c}{2} \frac{c}{n}\right)$. Raising both sides to the power of n and applying Proposition 24 gives the desired conclusion. $\square$

C Variation Distance between GMMs

Proof of Proposition 3: We use $d_{\text{TV}}(P, Q)$ and $\frac{1}{2}\|P - Q\|_1$ interchangeably in the cases where P and Q are not necessarily probability distributions. Let $\mathcal{N}_i = \mathcal{N}(\mu_i, \sigma_i^2)$ and $\hat{\mathcal{N}}_i = \mathcal{N}(\hat{\mu}_i, \hat{\sigma}_i^2)$. By triangle inequality,

$$d_{\text{TV}}(\hat{w}\hat{\mathcal{N}}_1 + (1 - \hat{w})\hat{\mathcal{N}}_2, w\mathcal{N}_1 + (1 - w)\mathcal{N}_2) \leq d_{\text{TV}}(\hat{w}\hat{\mathcal{N}}_1, w\mathcal{N}_1) + d_{\text{TV}}((1 - \hat{w})\hat{\mathcal{N}}_2, (1 - w)\mathcal{N}_2)$$

Inspecting the first term,

$$\frac{1}{2} \|w\mathcal{N}_1 - \hat{w}\hat{\mathcal{N}}_1\|_1 = \frac{1}{2} \|w\mathcal{N}_1 - w\hat{\mathcal{N}}_1 + w\hat{\mathcal{N}}_1 - \hat{w}\hat{\mathcal{N}}_1\|_1 \leq wd_{\text{TV}}(\mathcal{N}_1, \hat{\mathcal{N}}_1) + \frac{1}{2}|w - \hat{w}|,$$

again using the triangle inequality. A symmetric statement holds for the other term, giving us the desired result. $\square$

D Proofs Omitted from Section 3

Proof of Proposition 6: The probability that a sample is from $\mathcal{N}_i$ is w_i . Using the CDF of the half-normal distribution, given that a sample is from $\mathcal{N}_i$, the probability that it is at a distance $\leq \varepsilon\sigma_i$ from μ_i is $\text{erf}\left(\frac{\varepsilon}{\sqrt{2}}\right)$.

If we take a single sample from the mixture, it will satisfy the desired conditions with probability at least

$$w_i \text{erf}\left(\frac{\varepsilon}{\sqrt{2}}\right)$$

If we take $\frac{5\sqrt{\pi}}{w_i\varepsilon}$ samples from the mixture, the probability that some sample satisfies the conditions is at least

$$1 - \left(1 - w_i \text{erf}\left(\frac{\varepsilon}{\sqrt{2}}\right)\right)^{\frac{5\sqrt{\pi}}{w_i\varepsilon}} \geq 1 - e^{-5} \geq \frac{99}{100}$$

where the first inequality is Proposition 25. $\square$

Proof of Proposition 7: $w_i \geq \varepsilon$ implies $w_i \geq \frac{w_i + \varepsilon}{2}$, and thus $\frac{2}{\hat{w}_i} \geq \frac{2}{w_i + \varepsilon} \geq \frac{1}{w_i}$. $\square$

Proof of Lemma 8: Aside from the size of the collection, the rest of the conclusions follow from Propositions 6 and 7.

For a given $\hat{w}$, the number of pairs $(\hat{\mu}_1, \hat{\mu}_2)$ we consider can be written as

$$\frac{10\sqrt{\frac{\pi}{2}}}{\hat{w}\varepsilon} \cdot \frac{10\sqrt{\frac{\pi}{2}}}{(1 - \hat{w})\varepsilon} = \frac{50\pi}{\hat{w}(1 - \hat{w})\varepsilon^2}$$

We sum this over all candidates for $\hat{w}$, namely, $\varepsilon, 2\varepsilon, \dots, 1 - \varepsilon$, giving us

$$\sum_{k=1}^{\frac{1}{\varepsilon}-1} \frac{50\pi}{k\varepsilon(1 - k\varepsilon)\varepsilon^2}$$

Noting that this sum is symmetric about $k = \frac{1}{2\varepsilon}$, it reduces to

$$\sum_{k=1}^{\frac{1}{2\varepsilon}} \frac{100\pi}{k\varepsilon(1-k\varepsilon)\varepsilon^2} \leq \sum_{k=1}^{\frac{1}{2\varepsilon}} \frac{200\pi}{k\varepsilon^3} = \frac{200\pi}{\varepsilon^3} \sum_{k=1}^{\frac{1}{2\varepsilon}} \frac{1}{k} = \frac{200\pi}{\varepsilon^3} H_{\frac{1}{2\varepsilon}} = O\left(\frac{\log \varepsilon^{-1}}{\varepsilon^3}\right)$$

where H_n is the n th harmonic number. $\square$

Proof of Proposition 9: We will prove $\Pr(Y \geq \frac{c_2\sigma}{n}) \leq \frac{1}{20}$ and $\Pr(Y \leq \frac{c_1\sigma}{n}) \leq \frac{1}{20}$ separately. Using the union bound, this implies the desired statement.

Using the CDF of the half-normal distribution, the probability that $|X_i - \mu| < \frac{c_2\sigma}{n}$ is $\text{erf}\left(\frac{c_2}{\sqrt{2n}}\right)$. The probability that $Y > \frac{c_2\sigma}{n}$ is $\left(1 - \text{erf}\left(\frac{c_2}{\sqrt{2n}}\right)\right)^n = \left(\text{erfc}\left(\frac{c_2}{\sqrt{2n}}\right)\right)^n$. Set $c_2 = \sqrt{\frac{\pi}{2}} \ln 20$. Using Proposition 25,

$$\begin{aligned} \Pr\left(Y \geq \frac{c_2\sigma}{n}\right) &= \left(\text{erfc}\left(\frac{c_2}{\sqrt{2n}}\right)\right)^n \\ &\leq \exp(-\ln 20) \\ &= \frac{1}{20} \end{aligned}$$

Now, we must prove a bound in the opposite direction. The probability that $Y < \frac{c_1\sigma}{n}$ is $1 - \left(1 - \text{erf}\left(\frac{c_1}{\sqrt{2n}}\right)\right)^n = 1 - \left(\text{erfc}\left(\frac{c_1}{\sqrt{2n}}\right)\right)^n$. Set $c_1 = \sqrt{\frac{\pi}{2}} \frac{1}{25}$. Using Proposition 26,

$$\begin{aligned} \Pr\left(Y \leq \frac{c_1\sigma}{n}\right) &= 1 - \left(\text{erfc}\left(\frac{c_1}{\sqrt{2n}}\right)\right)^n \\ &\leq 1 - \left(1 - \frac{1}{25n}\right)^{\frac{1}{25}} \exp\left(-\frac{1}{25}\right) \\ &\leq \frac{1}{20} \end{aligned}$$

Where the last inequality holds if $n \geq 1$.

Using the union bound, the probability that Y lies outside the desired interval is $\leq \frac{1}{20} + \frac{1}{20} = \frac{1}{10}$, and we are done. $\square$

Proof of Proposition 11: Let X be the nearest sample from $\mathcal{N}_2$ to $\hat{\mu}_1$, and set $c_3 = \frac{1}{720\sqrt{\frac{2}{\pi}}}$. This proposition will hold if no sample from $\mathcal{N}_2$ lands in a window of width $2c_3\varepsilon\sigma_1$ centered at $\hat{\mu}_1$. The maximum value of $\mathcal{N}_2$'s PDF is $\frac{1}{\sigma_2\sqrt{2\pi}}$. As before, using standard Chernoff bounds, we have $\leq \frac{72w_2}{w_1\varepsilon}$

samples from $\mathcal{N}_2$ with probability at least $1 - \exp(-\frac{4w_2}{w_1\varepsilon}) \geq \frac{49}{50}$. Assuming this holds,

$$\begin{aligned}
& \Pr(|X - \hat{\mu}_1| \leq c_3\varepsilon\sigma_1) \\
& \leq 1 - \left(1 - 2c_3\varepsilon\sigma_1 \frac{1}{\sigma_2\sqrt{2\pi}}\right)^{\frac{72w_2}{w_1\varepsilon}} \\
& = 1 - \left(1 - \frac{1}{10} \frac{\varepsilon\sigma_2}{72\sigma_1}\right)^{\frac{72w_2}{w_1\varepsilon}} \\
& \leq 1 - \left(1 - \frac{1}{10} \frac{\varepsilon w_1}{72w_2}\right)^{\frac{72w_2}{w_1\varepsilon}} \\
& \leq 1 - e^{-\frac{1}{10}} \left(1 - \frac{1}{10} \frac{\varepsilon w_1}{72w_2}\right)^{\frac{1}{10}} \\
& \leq 1 - e^{-\frac{1}{10}} \left(1 - \frac{1}{10} \frac{1}{72}\right)^{\frac{1}{10}} \leq \frac{1}{10}
\end{aligned}$$

Therefore, $\Pr(|X - \hat{\mu}_1| \geq c_3\varepsilon\sigma_1) \geq \frac{49}{50} \cdot \frac{9}{10} \geq \frac{22}{25}$, as desired. $\square$

Proof of Lemma 12: Let Y be the nearest sample to $\hat{\mu}_1$. From Propositions 10 and 11, with probability $\geq \frac{7}{10}$, $|Y - \hat{\mu}_1| \in [\min\{(\frac{c_1}{36} - c_m), c_3\}\varepsilon\sigma_1, (\frac{c_2}{24} + c_m)\varepsilon\sigma_1]$.

We can generate candidates by rearranging the bounds to obtain

$$\frac{Y}{(\frac{c_2}{24} + c_m)\varepsilon} \leq \sigma_1 \leq \frac{Y}{\min\{(\frac{c_1}{36} - c_m), c_3\}\varepsilon}.$$

Applying Fact 22 and noting that $\frac{R}{L} = O(1)$, we conclude that we can grid over this range with $O(\frac{1}{\varepsilon})$ candidates. $\square$

Proof of Proposition 13: First, we show $E[Y] \leq \sigma\sqrt{2}\sqrt{\log n}$.

$$\begin{aligned}
\exp\{tE[Z]\} & \leq E[\exp\{tZ\}] \\
& = E[\max \exp\{tX_i\}] \\
& \leq \sum_{i=1}^n E[\exp\{tX_i\}] \\
& = n \exp\{t^2\sigma^2/2\}
\end{aligned}$$

The first inequality is Jensen's inequality, the second is the union bound, and the final equality follows from the definition of the moment generating function.

Taking the logarithm of both sides of this inequality, we get

$$E[Z] \leq \frac{\log n}{t} + \frac{t\sigma^2}{2}$$

This can be minimized by setting $t = \frac{\sqrt{2\log n}}{\sigma}$, which gives us the desired result,

$$E[Z] \leq \sigma\sqrt{2}\sqrt{\log n}$$

Next, we show the more difficult direction - that $E[Y] \geq \sigma C \sqrt{\log n}$, where $C = \frac{1}{\sqrt{\pi \log 2}}$. If one of our n samples is greater than some value, then the maximum is at least that value. Thus, we lower bound the expectation of Y by showing that we expect at least one of the X_i to be greater than $\sigma C \sqrt{\log n}$.

Let $k = \frac{2C^2}{\pi}$.

$$\begin{aligned}
E[|\{i : X_i \geq \sigma C \sqrt{\log n}\}|] &= nP(X_i \geq \sigma C \sqrt{\log n}) \\
&= \frac{n}{2}P(|X_i| \geq \sigma C \sqrt{\log n}) \\
&= \frac{n}{2} \left(1 - \operatorname{erf}\left(\frac{C}{\sqrt{2}} \sqrt{\log n}\right)\right) \\
&\geq \frac{n}{2} \left(1 - \sqrt{1 - \exp(-k \log n)}\right) \\
&= \frac{n}{2} \left(1 - \sqrt{1 - n^{-k}}\right)
\end{aligned}$$

Where the second equality follows by symmetry, the third equality is based on the CDF of the half-normal distribution, and the inequality is from the bound on the error function, $\operatorname{erf}(x) \leq \sqrt{1 - \exp(-\frac{4}{\pi}x^2)}$ [Wil46]. We require this value to be at least 1:

$$\begin{aligned}
\frac{n}{2} \left(1 - \sqrt{1 - n^{-k}}\right) &\geq 1 && \Leftrightarrow \\
1 - \frac{2}{n} &\geq \sqrt{1 - n^{-k}} && \Leftrightarrow \\
1 - \frac{4}{n} + \frac{4}{n^2} &\geq 1 - \frac{1}{n^k} && \Leftrightarrow \\
n^{2-k} &\geq 4n - 4 && \Leftrightarrow \\
(2 - k) \log n &\geq \log(4n - 4) && \Leftrightarrow \\
2 - \frac{\log(4n - 4)}{\log n} &\geq k
\end{aligned}$$

Substituting in the value $C = \frac{1}{\sqrt{\pi \log 2}}$ tell us this inequality holds for $n \geq 5.27$. Since we are only concerned with integer values of n , this inequality holds for $n \geq 6$. We can confirm the inequality for $1 \leq n \leq 5$ by comparing $\sigma C \sqrt{\log n}$ with the explicit equations for $E[Y]$ [Wei]. Note that this inequality is tight for $n = 2$. Therefore, we have $E[Y] \geq \sigma \frac{1}{\sqrt{\pi \log 2}} \sqrt{\log n}$. $\square$

Proof of Proposition 15: Using standard Chernoff bounds, with probability $\geq 1 - 2 \exp\left(-\frac{2^{10\pi+4}w_i}{12\varepsilon^2}\right) \geq 1 - 2 \exp\left(-\frac{2^{10\pi+2}}{3\varepsilon}\right)$, the number of samples from $\mathcal{N}_i$ is in the range $[\frac{w_i n}{2}, \frac{3w_i n}{2}]$. Using Proposition 14, with probability $\geq 1 - \frac{2}{e^{\frac{5}{2}}}$, the maximum sample lies within $\sqrt{5}\sigma_i$ of its expected value. These events both happen with probability $\geq \frac{4}{5}$. By taking the extreme points of both these cases and the inequalities in Proposition 13, we obtain the desired interval. $\square$

Proof of Proposition 16: Using a union bound, the conditions for Proposition 15 holds for both

Gaussians with probability $\geq \frac{3}{5}$. Thus, we must show that the two intervals are disjoint, that is,

$$\begin{aligned}
\mu_1 + \sigma_1 \left(\sqrt{2} \sqrt{\log \frac{3w_1 n}{2}} + \sqrt{5} \right) &\leq \mu_2 + \sigma_2 \left(\frac{1}{\sqrt{\pi \log 2}} \sqrt{\log \frac{w_2 n}{2}} - \sqrt{5} \right) && \Leftarrow \\
\sigma_1 \left(\sqrt{2} \sqrt{\log \frac{3w_1 n}{2}} + \sqrt{5} \right) &\leq \sigma_2 \left(\frac{1}{\sqrt{\pi \log 2}} \sqrt{\log \frac{w_2 n}{2}} - \sqrt{5} \right) && \Leftarrow \\
\sigma_1 \left(\sqrt{2} \sqrt{\log \frac{3 \cdot 2^{10\pi+4}}{2\varepsilon^2}} + \sqrt{5} \right) &\leq \sigma_2 \left(\frac{1}{\sqrt{\pi \log 2}} \sqrt{\log \frac{2^{10\pi+4}}{2\varepsilon}} - \sqrt{5} \right) && \Leftarrow \\
\left(\sqrt{2} \sqrt{\log \frac{3 \cdot 2^{10\pi+4}}{2\varepsilon^2}} + \sqrt{5} \right) &\leq C_2 \left(\frac{1}{\sqrt{\pi \log 2}} \sqrt{\log \frac{2^{10\pi+4}}{2\varepsilon}} - \sqrt{5} \right) && \Leftarrow \\
\frac{\left(\sqrt{2} \sqrt{\log \frac{3 \cdot 2^{10\pi+4}}{2\varepsilon^2}} + \sqrt{5} \right)}{\left(\frac{1}{\sqrt{\pi \log 2}} \sqrt{\log \frac{2^{10\pi+4}}{2\varepsilon}} - \sqrt{5} \right)} &\leq C_2
\end{aligned}$$

which is true for all $\varepsilon \leq 1$ if $C_2 \geq 9$.

In order to raise the probability of success to $\frac{19}{20}$, we use a standard boosting technique. We repeat this process several times and obtain the maximum sample from each run, then take the median of these maximums. In general, to raise the probability of success to $1 - \delta$, we require $O(\log \delta^{-1})$ repetitions. Since $\delta = \frac{1}{20}$, this will be a constant in this case. $\square$

Proof of Lemma 17: We can generate candidates for both cases, whether $\sigma_2 < C_2 \sigma_1$ or vice-versa. Since one of these is guaranteed to be true and both collections are of size $O(\frac{\log \varepsilon^{-1}}{\varepsilon})$, we only need to examine the two cases separately and then take the union of the two collections.

As described in Section 3.4.1, we can use a grid in the case where $\sigma_2 < C_2 \sigma_1$.

For the case where $\sigma_2 \geq C_2 \sigma_1$, apply Proposition 16. Again, we generate candidates for both cases, $\mu_2 \geq \mu_1$ and $\mu_2 \leq \mu_1$, where the statistics will be the maximum sample and the minimum sample, respectively, and take the union of the two. We examine the case where $\mu_2 \geq \mu_1$, the other case follows symmetrically.

Let Y be the maximum sample statistic we extract. Accounting for uncertainties in our parameters (noting that $\hat{w}_2 \in [\frac{w_2}{2}, 2w_2]$), we know that

$$Y \in \left[\hat{\mu}_2 + \sigma_2 \left(\frac{1}{\sqrt{\pi \log 2}} \sqrt{\log \frac{\hat{w}_2 n}{4}} - \sqrt{5} - 1 \right), \hat{\mu}_2 + \sigma_2 \left(\sqrt{2} \sqrt{\log 3\hat{w}_2 n} + \sqrt{5} + 1 \right) \right]$$

Rearranging the bounds, we can obtain

$$\frac{Y - \hat{\mu}_2}{\sqrt{2} \sqrt{\log 3\hat{w}_2 n} + \sqrt{5} + 1} \leq \sigma_2 \leq \frac{Y - \hat{\mu}_2}{\frac{1}{\sqrt{\pi \log 2}} \sqrt{\log \frac{\hat{w}_2 n}{4}} - \sqrt{5} - 1}$$

Applying Fact 22 and noting that $\frac{R}{L} = O(1)$, we conclude that we can grid over this range with $O(\frac{1}{\varepsilon})$ candidates. $\square$

D.1 Proof of Theorem 5

Proof of Theorem 5: We produce two lists of candidates corresponding to whether $\min(w, 1 - w) = \Omega(\varepsilon)$ or not:

- In the first case, combining Lemmas 8, 12, and 17 and taking the Cartesian product of the resulting candidates for the mixture's parameters, we see that we can obtain a collection of $O\left(\frac{\log^2 \varepsilon^{-1}}{\varepsilon^5}\right)$ candidate mixtures. With probability $\geq \frac{3}{5}$, this will contain a candidate $(\hat{w}, \hat{\mu}_1, \hat{\mu}_2, \hat{\sigma}_1, \hat{\sigma}_2)$ such that $\hat{w} \in w \pm c_w \varepsilon$, $\hat{\mu}_i \in \mu_i \pm c_m \varepsilon \sigma_i$ for $i = 1, 2$, and $\hat{\sigma}_i \in (1 \pm c_s \varepsilon) \sigma_i$ for $i = 1, 2$.

In order to obtain the desired total variation distance from the unknown distribution, as suggested by Lemma 4, we must set $c_w \leq \frac{1}{3}$, $c_m \leq \frac{1}{4}$, and $c_s \leq \frac{1}{8}$. In particular, note that we must set c_m sufficiently small to satisfy the conditions of Lemma 12.

Finally, note that the number of samples that we need for the above to hold is $O(1/\varepsilon^2)$. For this, it is crucial that we *first* draw a sufficient $O(1/\varepsilon^2)$ samples from the mixture (specified by the worse requirement among Lemmas 8, 12, and 17), and *then* execute the candidate generation algorithm outlined in Lemmas 8, 12, and 17. In particular, we do not want to redraw samples for every branching of this algorithm.

- In the second case, i.e. when one of the weights, w.l.o.g. w_2 , is $O(\varepsilon)$, we set $\hat{w}_1 = 1$ and we only produce candidates for (μ_1, σ_1^2) . This case follows similarly, and amounts to applying Lemma 8 and Proposition 9. We first run the procedure suggested by Lemma 8 to obtain an estimate of μ_1 , discarding the candidates for w, μ_2 . We then draw a single sample from the mixture - with probability $1 - O(\varepsilon)$, this will correspond to a sample from $\mathcal{N}_1$. Applying Proposition 9 for the case $n = 1$, with probability $\geq \frac{9}{10}$, this sample will lie at a distance $[(c_1 - c_m \varepsilon) \sigma_1, (c_2 + c_m \varepsilon) \sigma_1]$ from the estimated mean $\hat{\mu}_1$. This provides us with an interval we can grid over to obtain a collection of $O(\frac{1}{\varepsilon})$ candidates for σ_1 such that at least one $\hat{\sigma}_1 \in (1 \pm \varepsilon) \sigma_1$. Again, the success probability is $\geq 1 - \frac{1}{50} - \frac{1}{10} - O(\varepsilon) \geq \frac{3}{5}$ for ε sufficiently small. Similar to the first case, we set $c_w \leq \frac{1}{3}$, $c_m \leq \frac{1}{4}$, $c_s \leq \frac{1}{8}$ in order to satisfy the conditions of Lemma 4, and draw $O(1/\varepsilon^2)$ samples from the mixture before we begin.

Finally, to boost the success probability, we repeat the entire process $\log_{\frac{5}{2}} \delta^{-1}$ times and let our collection of candidate mixtures be the union of the collections from these repetitions. The probability that none of these collections contains a suitable candidate distribution is $\leq \left(\frac{2}{5}\right)^{\log_{\frac{5}{2}} \delta^{-1}} \leq \delta$. $\square$

E Proofs Omitted from Section 4

Proof of Claim 1: Let $\tau = X(\mathcal{W}_1)$. The Chernoff bound implies that, with probability at least $1 - 6e^{-m\varepsilon^2/2}$, the following are simultaneously true: $|p_1 - \hat{p}_1| < \varepsilon/2$, $|p_2 - \hat{p}_2| < \varepsilon/2$, $|\tau - \hat{\tau}| < \varepsilon/2$. Conditioning on these:

- If $d_{TV}(X, H_1) \leq \varepsilon$ and $d_{TV}(X, H_2) > 8\varepsilon$, then from the triangle inequality we get that $p_1 - p_2 = d_{TV}(H_1, H_2) > 7\varepsilon$, hence $\hat{p}_1 - \hat{p}_2 > p_1 - p_2 - \varepsilon > 6\varepsilon$. Hence, the algorithm will go beyond Step 2. Moreover, $d_{TV}(X, H_1) \leq \varepsilon$ implies that $|\tau - p_1| \leq \varepsilon$, hence $|\hat{\tau} - \hat{p}_1| < 2\varepsilon$. So the algorithm will stop at Step 3, declaring H_1 as the winner of the competition between H_1 and H_2 .

- If $d_{TV}(X, H_2) \leq \varepsilon$ and $d_{TV}(X, H_1) > 8\varepsilon$, then as in the previous case we get from the triangle inequality that $p_1 - p_2 = d_{TV}(H_1, H_2) > 7\varepsilon$, hence $\hat{p}_1 - \hat{p}_2 > p_1 - p_2 - \varepsilon > 6\varepsilon$. Hence, the algorithm will go beyond Step 2. Moreover, $d_{TV}(X, H_2) \leq \varepsilon$ implies that $|\tau - p_2| \leq \varepsilon$, hence $|\hat{\tau} - \hat{p}_2| < 2\varepsilon$. So $\hat{p}_1 > \hat{\tau} + 4\varepsilon$. Hence, the algorithm will not stop at Step 3, and it will stop at Step 4 declaring H_2 as the winner of the competition between H_1 and H_2 .
- If $d_{TV}(X, H_1) \leq \varepsilon$ and $d_{TV}(X, H_2) > 4\varepsilon$, we distinguish two subcases. If $\hat{p}_1 - \hat{p}_2 \leq 6\varepsilon$, then the algorithm will stop at Step 2 declaring a draw. If $\hat{p}_1 - \hat{p}_2 > 6\varepsilon$, the algorithm proceeds to Step 3. Notice that $d_{TV}(X, H_1) \leq \varepsilon$ implies that $|\tau - p_1| \leq \varepsilon$, hence $|\hat{\tau} - \hat{p}_1| < 2\varepsilon$. So the algorithm will stop at Step 3, declaring H_1 as the winner of the competition between H_1 and H_2 .
- If $d_{TV}(X, H_2) \leq \varepsilon$ and $d_{TV}(X, H_1) > 4\varepsilon$, we distinguish two subcases. If $\hat{p}_1 - \hat{p}_2 \leq 6\varepsilon$, then the algorithm will stop at Step 2 declaring a draw. If $\hat{p}_1 - \hat{p}_2 > 6\varepsilon$, the algorithm proceeds to Step 3. Notice that $d_{TV}(X, H_2) \leq \varepsilon$ implies that $|\tau - p_2| \leq \varepsilon$, hence $|\hat{\tau} - \hat{p}_2| < 2\varepsilon$. Hence, $\hat{p}_1 > \hat{p}_2 + 6\varepsilon \geq \hat{\tau} + 4\varepsilon$, so the algorithm will not stop at Step 3 and will proceed to Step 4. Given that $|\hat{\tau} - \hat{p}_2| < 2\varepsilon$, the algorithm will stop at Step 4, declaring H_2 as the winner of the competition between H_1 and H_2 .
- If $d_{TV}(H_1, H_2) \leq 5\varepsilon$, then $p_1 - p_2 \leq 5\varepsilon$, hence $\hat{p}_1 - \hat{p}_2 \leq 6\varepsilon$. So the algorithm will stop at Step 2 declaring a draw. $\square$